Introductory Econometrics

Introduction (Lecture 0)

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WiSe 2025/2026

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Outline

Formalities

Big picture

Course Outline

Data structures

Motivation

Causality

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Formalities

General remarks:

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    - Theory
    - Practice: data cleaning, R, models
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    - **Monday at 15.45**, by appointment (I will post details on Moodle)

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  - Monday, 26.01.2026
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- Tutorials to discuss problem sets:
  - Voluntary and not graded
  - First tutorial: Thursday 09.10.2025 13:15 - 14:45
  - Each Thursday

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Registration and deregistration

Registration:

- People who signed up online for this class and signed the list today are considered enrolled

- If you did not sign up online, or your name was not on the list
  - please add your name and student ID number manually
  - come see me after this class to discuss if you fulfill the requirements
  - send me an email (unless you have done that already)

Deregistration:

- If you are registered, you will receive a grade

- Deregistration possible until **Fr 14.10.2025 23:59**

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Textbooks

Useful textbooks:

- J. Wooldridge (2019), 7th edition: Introductory Econometrics, Cengage Learning.
- J. Stock and M.Watson (2019): Introduction to Econometrics, Pearson.
- J. Angrist and S. Pischke (2008): Mostly Harmless Econometrics, Princeton Univ. Press.

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Code used in this course will be provided in R and Stata. You are highly encouraged to look into these examples! All examples and illustrations on the slides are prepared in R and will be shared on demand

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- **RStudio** is freely available and can be downloaded under www.rstudio.com
- We will help you using R in the next tutorial

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Stata:

- **Stata** is not free, but can be accessed in the computer labs
- Student licences are available zid.univie.ac.at/software-fuer-studierende/anleitungen/stata/
- For Stata, the following sources are useful:
  - Germán Rodriguez Tutorial (Princeton University): data.princeton.edu/stata/
  - UCLA Stata Tutorials: stats.idre.ucla.edu/stata/
  - LSE Video Tutorials www.lse.ac.uk/Methodology/Software-tutorials/Statatutorials
  - Stata blogs: https://www.statalist.org/

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Python:

- Download **python** via Anaconda: https://www.anaconda.com/download
- You can use Spyder through Anaconda
- Alternatively, use it on the browser on Jupyter: https://jupyter.org/

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ChatGPT, MS Copilot, Claude, and other AI tools:

- They can help you in understanding and writing the code or solve errors you may encounter
- Use it **wisely**!
- **Do not** copy and paste, but understand the underlying problem (also for the homework!)

Outline

Big picture

What to expect from this course?

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- Basic goal: answer **economic questions** using **data**

$\rightarrow$ data structures

$\rightarrow$ econometrics

$\rightarrow$ required assumptions

$\rightarrow$ from correlation to causality

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Course outline I

1. Introduction
    - Motivation
    - The nature of econometric data, problems, and analyses (W, ch. 1)
    - Review of probability & statistics (W, appendix B-C)
2. Bivariate regression (W, ch. 2)
    - The simple regression model
    - OLS estimator
    - Basic properties of OLS
    - Variance estimation
3. Multivariate regression (W, ch. 3)
    - Matrix notation
    - Derivation and mechanics of OLS
    - Omitted variable bias
    - Gauss-Markov
    - Inference and testing (W, ch. 4)
    - Asymptotics (W, ch. 5)

Course outline II

4. Further topics in regression analyses
    - Functional form, dummy variables (W, ch. 6–7)
    - Testing
    - Heteroscedasticity and generalized error term structures (W, ch. 8)

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Table: Mock data on income of individuals

id	income	age
1	0	68
2	384	23
3	0	34
4	1795	32

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Table: GDP of Gambia

year	GDPpc	growth
2017	680	0.98
2018	733	1.08
2019	773	1.05
2020	757	0.98
2021	836	1.10

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Table: City populations over time

year	city	population
2018	Vienna	1,889,000
2019	Vienna	1,890,000
2020	Vienna	1,911,000
2018	Frankfurt/Main	769,000
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Tip – data structure

Learning to manipulate data and go from this:

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to this:

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and the opposite!

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$\rightarrow$ this is done using reshape in STATA or pivot in R

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- What will the unemployment rate be next quarter?

  Forecasting Q: asks what will happen in the future

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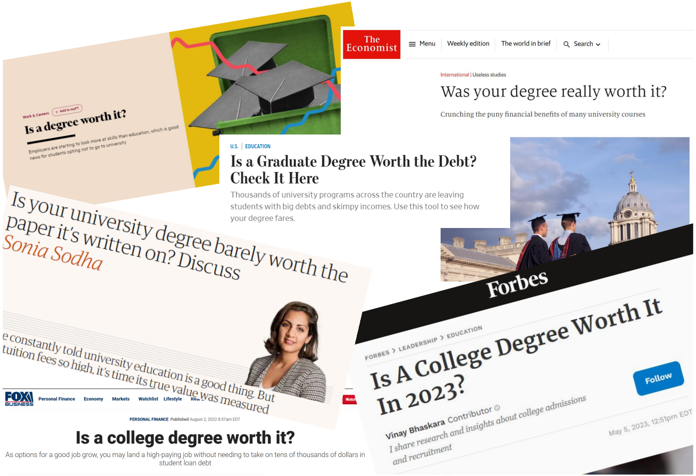
4. **prediction and forecasting**:
   - by how much will GDP grow next year?
   - how volatile will stock markets be next week?
   - if we offer microinsurance to farmers in rural Ethiopia, how many will enroll?

Too many university students?

You are an economist working for the government



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Putting Econometrics at work

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- A dependent variable
- An independent variable

It does not seem an easy task!

Putting Econometrics at work

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And, as economists, we might want to evaluate the effect of college education on wages, to show that there are significant **“returns” to college investments**

- College premium

- College return

A simple model

First, develop a model of **wage** determination for individual i, y_i, as a function of observed characteristics, $x_{1i}, \ldots, x_{ki}$, and unobserved characteristics, $u_{1i}, \ldots, u_{li}$:

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A very generic model

$$y_i = f(x_{1i}, \ldots, x_{ki}, u_{1i}, \ldots, u_{ki})$$

- $x_{1i}, \ldots, x_{ki}$ are variables that are **observed** in the data
- $u_{1i}, \ldots, u_{li}$ are variables that are **not observed** in the data

candidates for

- **observed variables** $(x_{2i}, \ldots, x_{ki})$:

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- **unobserved variables** $(u_{1i}, \ldots, u_{li})$:
  - general ability
  - specific ability
  - motivation
  - diligence

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  - $Y_i(1) - Y_i(0)$: individual i's "treatment effect" (returns to college)

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- Observed earnings outcome Y_i is $Y_i(1)$ if $D_i = 1$ and $Y_i(0)$ if $D_i = 0$: $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$

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    - We set our target parameter as the average effect: $\mathbb{E}[Y_i(1) - Y_i(0)]$

Let's Get Specific (Cont.)

Suppose we have a **sample** of earnings and schooling: (Y_i, D_i), $i = 1, ..., N$

- Take as our **estimator** the difference in sample average earnings for the $N1$ people who did go to college vs. the $N0$ people who didn't go to college:

$$\frac{1}{N1} \sum_{i:D_i=1} Y_i - \frac{1}{N0} \sum_{i:D_i=0} Y_i$$

- Our **estimand** is the corresponding difference in population means:

$$\mathbb{E}[Y_i | D_i = 1] - \mathbb{E}[Y_i | D_i = 0]$$

- Compare this to our **parameter**: average treatment effect $E[Y_i(1) - Y_i(0)]$

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From Estimator to Estimand

Our first task is to learn about unobserved population means $\mathbb{E}[Y_i|D_i = d]$ from observed sample means $\frac{1}{N1} \sum_{i:D_i=d} Y_i$ (**statistical inference**)

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We will review all these concepts, but...

From Estimand to Parameter

Suppose we were somehow able to see the entire population: i.e. the avg. earnings of all people who did/didn't go to college $\mathbb{E}[Y_i | D_i = d]$

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- Note that under our model, $\mathbb{E}[Y_i | D_i = d] = \mathbb{E}[Y_i(d) | D_i = d]$

From Estimand to Parameter

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Selection bias: people who do/don't go to college (D_i) have different potential earnings with/without college ($Y_i(d)$):

- Differences in academic ability, family background, career goals, etc.
- Sometimes referred to as **omitted variables** or confounding factors

Identification challenge: Only observe $Y_i(1)$ among those with $D_i = 1$ and $Y_i(0)$ among those with $D_i = 0$; **counterfactual** $Y_i(d)$'s are unobserved

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When does selection arise? Role of π on θ

Model (latent-index selection):

$$D_i = 1[\alpha + \pi\theta_i + u_i > 0], \qquad Y_i(d) = \gamma_d + \lambda_d \theta_i + \varepsilon_i$$

Assumptions: $u_i \perp (\theta_i)$ and F strictly increasing.

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Propensity is monotone in θ:

$$p(x, \theta) \equiv \Pr(D_i = 1 \mid, \theta_i = \theta) = F(\alpha + \pi\theta)$$

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Positive selection on θ when $\pi > 0$:

$$\pi > 0 \;\Rightarrow\; \mathbb{E}[\theta_i \mid D_i = 1] \;>\; \mathbb{E}[\theta_i] \;>\; \mathbb{E}[\theta_i \mid D_i = 0]$$

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Implication for observed means (selection term in bold):

$$\mathbb{E}[Y_i(d) \mid D_i = d] = \gamma_d + \lambda_d \underbrace{\mathbb{E}[\theta_i \mid D_i = d]}_{\textbf{selection on } \theta}$$

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Therefore:

$$\mathbb{E}[Y_i(d) \mid D_i = d] - \mathbb{E}[Y_i(d)] = \lambda_d(\mathbb{E}[\theta_i \mid D_i = d] - \mathbb{E}[\theta_i])$$

$$\pi \neq 0 \;\Rightarrow\; \mathbb{E}[\theta_i \mid D_i = d] \neq \mathbb{E}[\theta_i] \quad \Longrightarrow \quad \mathbb{E}[Y_i(d) \mid D_i = d] \neq \mathbb{E}[Y_i(d)].$$

If $\pi > 0$ *and* $\lambda_d > 0$, *the bias is upward for* $D_i = 1$ (positive selection).

Selection bias

- **Selection bias**: people who do/don't go to college (D_i) have different potential earnings with/without college ($Y_i(d)$):
  - Differences in academic ability, family background, career goals, etc. (θ)
  - Sometimes referred to as **omitted variables** or confounding factors (omitting to include θ)
  - On average, people with higher ability will receive a college diploma with higher probability (*pi*) and earn higher potential wages (in both counterfacutals: λ_d)

Identification through Randomization

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- But randomization is often **hard/impossible/unethical** in economics
  - E.g. imagine convincing states to randomize college attendance!
  - We will develop tools to “mimic” the RCT gold standard, in non-experimental (“observational”) data

Outline

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what kind of **statistical experiment** could we design to infer such a causal relationship?

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data can be obtained from a variety of sources/methods/settings

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- in this course we focus on “ordinary least squares estimation”
  - learn when it is valid, and when it is not
  - what assumptions are needed

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Ex.: The causal effect of class size on test scores

Suppose, we study the **average causal effect** of changing district-level class sizes on student test scores. An **RCT** would have to:

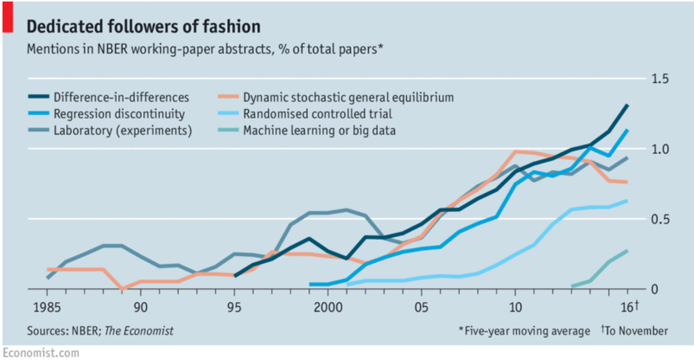
- randomly reallocate teachers to districts to vary class sizes
- prohibit students and teachers from changing schools
- and give identical tests to students across states

Popularity of experiments

- e.g., in development econ, labor econ, health econ

Focus on the lines for

- **Laboratory** and
- **Randomized control trials**



Source: economist.com/finance-and-economics/2016/11/24/economists-are-prone-to-fads-and-the-latest-is-machine-learning

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advantages:

- often much larger in **scope** and **scale**. May comprise the entire population of a country
- typically reflects **real-life behavior** of individuals, households, firms, rather than artificial experimental setting (cf. field experiments)

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Because **observational data** is prevalent in social sciences, this class focus on this type and study **assumptions and situations** under which we can **learn about causal effects** even with observational data.

Recap

- data structures (cross-sectional, time series, panel)
- types of questions to be answered using econometrics:
  - falsifying theories, quantifying relationships, treatment evaluation, forecasting
- example of wages and education:
  - model, visual analysis, basic quantitative analysis
  - issues: Estimation uncertainty, causality
- causality

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